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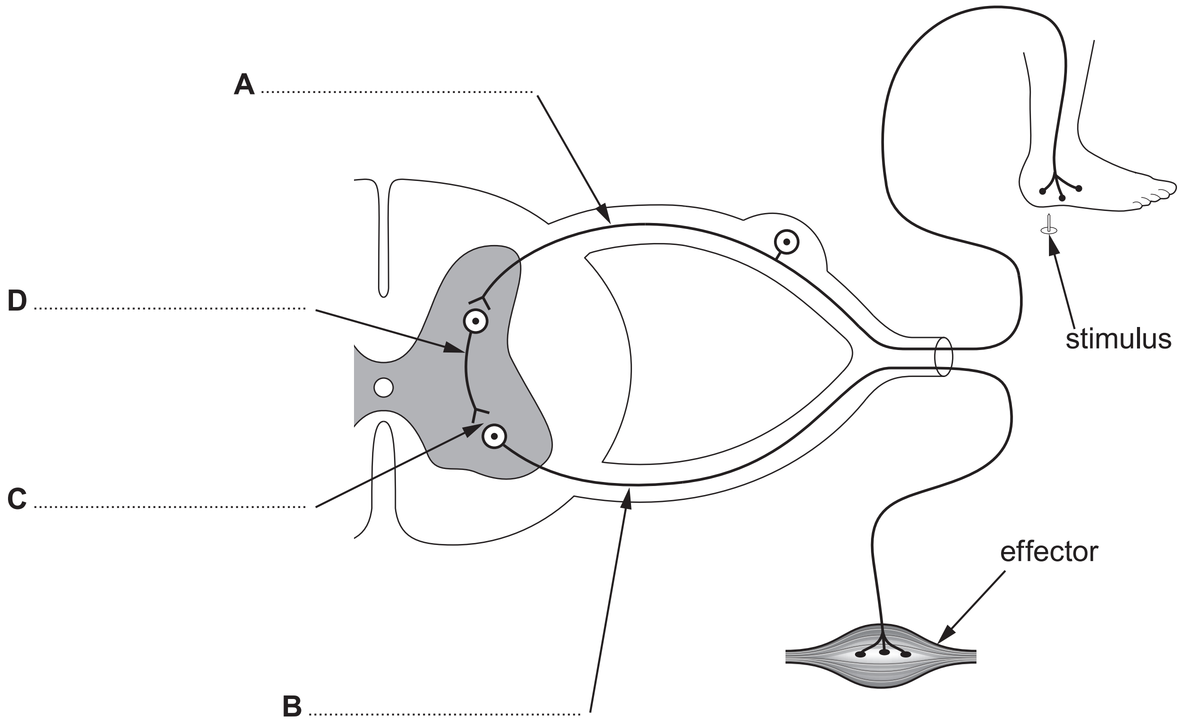
4. **Image 4.1** shows a reflex arc.



(a) (i) Label **A–D** on **Image 4.1**. [4]

(ii) State **two** properties of reflex actions. [1]

....

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(iii) The length of neurone **B** is 0.9 m. An electrical impulse can travel along a neurone at 75 m/s.

Use the following equation:

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

to calculate the time taken for the impulse to travel the length of neurone **B**. [2]

Time = ...s

(b) Motor neurone disease prevents motor neurones functioning effectively.
Explain why individuals with the disease find it difficult to walk. [2]

....

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